Paddy Prediction

The training set has images **and the following metadata**

```
Loaded CSV: data/train.csv (rows=10407, cols=4)
```

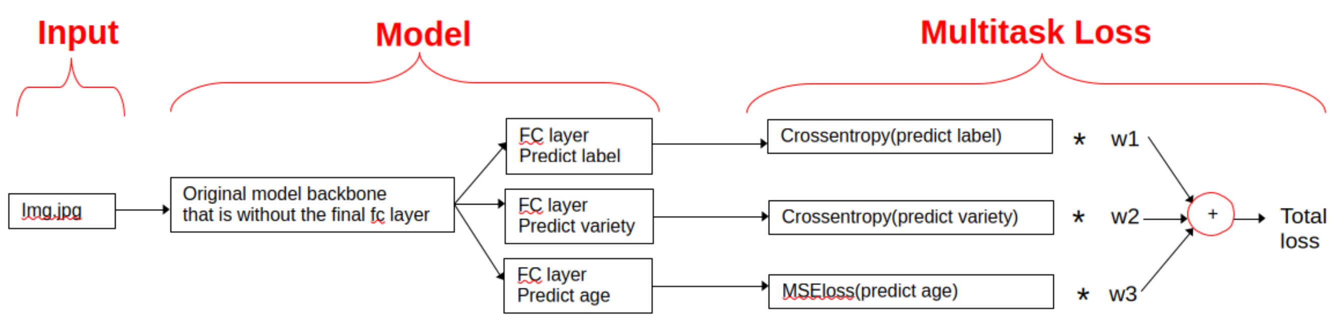
	image_id	label	variety	age
0	100330.jpg	bacterial_leaf_blight	ADT45	45
1	100365.jpg	bacterial_leaf_blight	ADT45	45
2	100382.jpg	bacterial_leaf_blight	ADT45	45
3	100632.jpg	bacterial_leaf_blight	ADT45	45
4	101918.jpg	bacterial_leaf_blight	ADT45	45

The submission set (in /test_images), has **just an image, no metadata**

Training Overview- when you have meta data during train but NOT test

Models often perform better on everything when number of tasks increases. So get the model to predict the metadata during training with only the image as <u>input</u> to the model (so can still inference with just image). Instead of 1 head, add 3. W1,W2 and W3 weight each loss.

Multi <u>Head</u> Model – multiple training heads and custom loss function



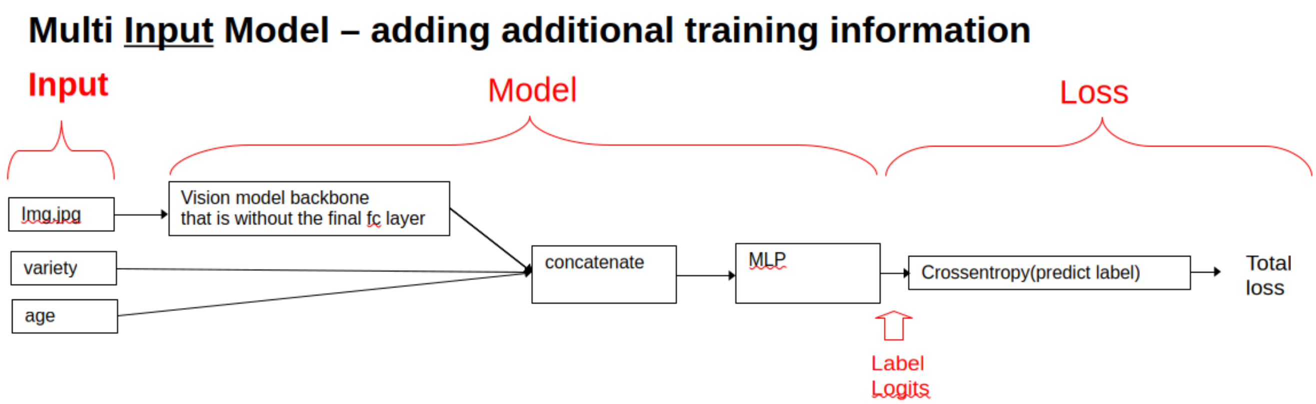
<u>Need</u>

- Dataset that can deliver the image <u>and</u> label, variety and age metadata
- model with 3 heads
- criterion (loss function) that can do loss on each head and produce a weighted sum
- plus various changes to code (llms will automatically do this)

Strategy:

train the FC layers first with the rest of the model frozen
then unfreeze entire model and train some more at lr/10 (this is where model will ingest new information that comes from it predicting age and variety)

Training Overview- when you have meta data during both train and test
If metadata helps then feed it into model. Note that it is concatenated to the vision models output logits which are then fed into a MLP to predict 1 of 10 labels



Need
- Dataset from above
- model as above
- various changes to code (llms will automatically do this)

Strategy:
train the FC layers first with the rest of the model frozen
then unfreeze entire model and train some more at lr/10